The Bellman Residual Is Not a Policy-Improvement Proxy in Actor–Critic Reinforcement Learning

Anonymous Author(s)
Affiliation
Address
email

Abstract

Actor–critic deep reinforcement learning (RL) relies on learned value functions to guide policy updates. Practitioners typically monitor the Bellman residual (ϵ_u), or value-function loss, as a proxy for critic quality and update health. We demonstrate, both theoretically and empirically, that the absolute Bellman residual is structurally uninformative as a per-update predictor of policy improvement. Theoretically, we show that the relevant criterion is a *dimensionless ratio* ϵ_u/A_k, and that any fixed absolute tolerance eventually becomes vacuous as the policy improves. Empirically, we audit over 86,000 actor–critic updates across PPO, SAC, and TRPO, spanning eight environments and 237 seeds. We find ϵ_u is near-chance at predicting policy harm (pooled median AUROC 0.46). The exact policy-improvement identity $J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k$ exposes the **start-state critic bias** $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$ as the operative covariate-shift quantity. The bias term predicts harm at pooled median AUROC 0.68 and dominates ϵ_u on 234 of 236 paired seeds (paired Wilcoxon $p < 10^{-40}$). To rule out return-estimate leakage, we replicate the diagnostic on independent held-out rollouts and recover AUROC 0.61, confirming the signal reflects critic generalization rather than shared Monte Carlo noise. The dominance is uniform across PPO, SAC, and TRPO, robust to a ~ 200-cell hyperparameter factorial, and matches a tabular sanity check where the identity holds to machine precision. We conclude that critic monitoring and critic-objective design in actor–critic deep RL should shift from absolute TD error toward relative, distribution-aware quantities.

1 Introduction

Actor–critic algorithms in deep reinforcement learning (RL)—including PPO [Schulman et al., 2017], SAC [Haarnoja et al., 2018], and TRPO [Schulman et al., 2015a]—update a policy iteratively against a learned value function. The standard analytic guarantee for the success of these updates typically relies on a bound of the form:

$$J(\pi_{k+1}) - J(\pi_k) \;\geq\; A_k \;-\; c \cdot \epsilon_u, \tag{1}$$

where ϵ_u is a norm of the Bellman residual $T^\pi V - V$ and A_k is the advantage signal. In practice, the scalar ϵ_u is the value-function loss minimized by nearly every deep actor–critic implementation [Huang et al., 2022, Andrychowicz et al., 2021], and is the primary quantity practitioners monitor to assess whether their critic is "good enough" to support policy improvement.

In this paper, we show that this reliance on the absolute Bellman residual is misplaced. We argue that ϵ_u is the wrong object for monitoring update health, both because it lacks the correct scale and because it targets the wrong distribution.

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Theory: absolute tolerances are structurally doomed. We first provide a structural argument for why absolute Bellman residuals cannot certify policy improvement. We show that the relevant theoretical tolerance is the *dimensionless ratio* ϵ_u/A_k. As training progresses and the policy approaches a local optimum, the advantage signal A_k naturally tends toward zero. Consequently, any fixed absolute tolerance ϵ^{fixed} eventually becomes too large relative to the available improvement signal, rendering the improvement guarantee (1) vacuous. We formalize this in Proposition 2, showing that fixed-tolerance critics eventually fail to certify even genuine improvements.

Mechanism: the start-state bias as covariate shift. The failure of the Bellman residual to track improvement is explained by an exact algebraic identity (Proposition 1):

$$J(\pi_{k+1}) - J(\pi_k) \;=\; A_k \;-\; \hat{\Delta}_k, \qquad \hat{\Delta}_k = J(\pi_k) - \mathbb{E}_{s\sim\nu}[V_{k+1}(s)]. \tag{2}$$

We call $\hat{\Delta}_k$ the **start-state critic bias**: it is the post-update critic's error *at the initial-state distribution* ν. The Bellman residual ϵ_u measures critic error on the rollout distribution d^{π_k}; it is silent about accuracy at ν. The two distributions disagree even on-policy because d^{π_k} is the discounted state-visitation distribution and ν is its starting point. The diagnostic failure is then mechanically clear: ϵ_u measures the right kind of error on the wrong distribution.

Empirical audit across PPO, SAC, and TRPO. We validate these theoretical insights with an audit of over 86,000 actor–critic updates spanning PPO, SAC, and TRPO across eight environments and 237 seeds. We find that the canonical Bellman residual is near-chance at predicting whether an update will be beneficial or harmful (pooled median AUROC 0.46). In contrast, the start-state bias $-\hat{\Delta}_k$ predicts harm at pooled median AUROC 0.68 and dominates ϵ_u on 234 of 236 paired seeds (paired Wilcoxon $p < 10^{-40}$). The dominance holds uniformly across all four algorithm variants we test (PPO with the standard clipped-MSE critic, PPO with a pure-MSE critic ablation, SAC, and TRPO), across discrete and continuous control, and across a $\sim$200-cell hyperparameter factorial.

Inoculations against the obvious objections. A natural concern is that $\widehat{\hat{\Delta}}_k$ and ΔJ_k share the episode return $\widehat{J}(\pi_k)$ in their definitions, raising the possibility of finite-sample leakage. We replicate the diagnostic on *fully independent* held-out rollouts (separate transitions and separate start states) and recover AUROC 0.61 on Humanoid-v5, confirming the signal reflects critic generalization. A second concern is that $\hat{\Delta}_k$ is post-hoc—it depends on the post-update critic V_{k+1}. We confirm the consequence: the lag-1 AUROC (using the metric at step k to predict harm at step $k{+}1$) collapses to 0.51, demarcating a clean empirical boundary between retrospective diagnostics and prospective controllers. We discuss this boundary as a *positive* contribution—it specifies what objective a future critic loss should target, even if $\hat{\Delta}_k$ itself cannot be used in real time.

Contributions. Our contributions are: (i) a structural argument that absolute Bellman residuals cannot certify policy improvement, formalized as a no-go theorem (Proposition 2); (ii) identification of the start-state critic bias $\hat{\Delta}_k$ as the mechanistically active signal, via the exact identity $J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k$; (iii) a large-scale audit across PPO, SAC, and TRPO giving the first cross-algorithm empirical evidence that value-function loss is decoupled from policy improvement.

2 Related work

Empirical studies of deep policy gradients. Our work builds on the critical evaluation of deep RL implementations. Henderson et al. [2018] and Engstrom et al. [2020] demonstrated that implementation details often dominate algorithmic choices. Most relevantly, Ilyas et al. [2020] established that critics fail to fit the true value function globally—the network solves the supervised task but not the estimation problem. Our work addresses a logically prior question: even accepting the critic quality produced by a run, is the standard scalar used to *measure* that quality (ϵ_u) informative about policy improvement? We identify the covariate shift between d^{π_k} and ν as the algebraic reason for this failure.

Value overestimation. The overestimation literature [Fujimoto et al., 2018] motivates critic design choices, such as clipped double-Q in SAC, by targeting overestimation on the rollout distribution.

Our results imply a sharper target: what matters for policy improvement is the bias at ν, not the average error over d^{π_k}. Our SAC audit confirms that while these methods stabilize training, they do not close the gap between ϵ_u and $\hat{\Delta}_k$ as diagnostic predictors.

Distribution shift in RL. Prior work on distribution shift has largely focused on the behavior/target policy mismatch in off-policy learning [Kakade and Langford, 2002, Islam et al., 2019]. We identify the ν vs. d^{π_k} mismatch as the operative gap even in on-policy methods like PPO. This reframes start-state bias as a fundamental covariate shift problem inherent to the actor–critic objective itself.

3 Audit setup

We audit four actor–critic configurations spanning three algorithm families.

Algorithms and environments. PPO [Schulman et al., 2017] on the canonical 8-environment suite at 200k–500k timesteps each: discrete-action LunarLander-v3, CartPole-v1, Acrobot-v1, and MuJoCo continuous control on Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5, Humanoid-v5; 20 seeds per environment (160 runs). PPO-MSE: an ablation that replaces PPO's clipped value loss with a pure mean-squared TD error, on the same 8 envs $\times$ 5 seeds. SAC [Haarnoja et al., 2018] on the three MuJoCo envs where it is known to train (≥ 10 seeds each). TRPO [Schulman et al., 2015a] on LunarLander-v3 and Hopper-v5 (≥ 9 seeds each). Three seeds stalled below 10 updates (one PPO CartPole seed plateaued at the 500-step solved threshold, one SAC Hopper, one TRPO LunarLander) and are excluded from per-seed AUROC; 237 seeds remain (Table 1, n column). Total: $\sim$86,000 actor–critic updates with paired (diagnostic, ground-truth ΔJ_k) records.

Logging schema. For each update we log $\widehat{A}_k$, $\widehat{\hat{\Delta}}_k$ (the start-state estimator, eq. (6)), $\widehat{\epsilon}_u$ (mean squared TD error on the rollout, post-critic-update, the canonical scalar minimized in PPO/SAC/TRPO codebases), and the ground-truth return $\widehat{J}(\pi_k)$. The harmful label is $1[\widehat{\Delta J}_k < 0]$, where $\widehat{\Delta J}_k = \widehat{J}(\pi_{k+1}) - \widehat{J}(\pi_k)$ from successive rollout batches. The harmful label is undefined on the final update of each run.

Hyperparameter factorials. To rule out tuning artifacts we run three factorial sweeps. LunarLander: 4×4 over value-epochs $\in \{5, 10, 20, 40\}$ and clip-$\epsilon \in \{0.1, 0.2, 0.3, 0.4\}$, 5 seeds per cell (80 runs). HalfCheetah and Hopper: 3×2 over value-epochs $\in \{5, 10, 20\} \times$ clip-$\epsilon \in \{0.1, 0.2\}$ plus a 3-cell value-learning-rate sweep at the default cell, 5 seeds each (45 runs per env). All factorials use the same logging schema. Full details in Section F.

4 The improvement identity and practical estimators

We consider an infinite-horizon discounted MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \nu)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, reward R, discount $\gamma \in [0, 1)$, and initial-state distribution $\nu \in \Delta(\mathcal{S})$. For a policy π, write V^π for its value function, $J(\pi) = \mathbb{E}_{s_0 \sim \nu}[V^\pi(s_0)]$ for its expected return, T^π for the Bellman operator, and d^π for the discounted state-visitation distribution from ν:

$$(T^\pi V)(s) = \mathbb{E}_{a\sim\pi}\big[R(s,a) + \gamma \mathbb{E}_{s'}[V(s')]\big], \qquad (d^\pi)^\top = (1-\gamma)\,\nu^\top (I - \gamma P^\pi)^{-1}.$$

Consider an iterative algorithm that, at iteration k, holds π_k and a critic estimate V_k, and produces (π_{k+1}, V_{k+1}). The estimate V_{k+1} need not coincide with $V^{\pi_{k+1}}$.

Proposition 1 (Improvement identity at the initial-state distribution)**.** *For any policies π_k, π_{k+1} and any value function $V_{k+1} : \mathcal{S} \to \mathbb{R}$,*

$$J(\pi_{k+1}) - J(\pi_k) \;=\; A_k \;-\; \hat{\Delta}_k, \tag{3}$$

where

$$A_k := \tfrac{1}{1-\gamma}\, \mathbb{E}_{s\sim d^{\pi_{k+1}}}[(T^{\pi_{k+1}} V_{k+1})(s) - V_{k+1}(s)] \text{ and } \hat{\Delta}_k := J(\pi_k) - \mathbb{E}_{s\sim\nu}[V_{k+1}(s)]. \tag{4}$$

Proof. The Bellman residual at π_{k+1} on V_{k+1} is $T^{\pi_{k+1}} V_{k+1} - V_{k+1} = r^{\pi_{k+1}} + \gamma P^{\pi_{k+1}} V_{k+1} - V_{k+1}$. Substituting $(d^{\pi_{k+1}})^\top = (1-\gamma)\nu^\top (I - \gamma P^{\pi_{k+1}})^{-1}$ gives

$$(1-\gamma) A_k = \nu^\top (I - \gamma P^{\pi_{k+1}})^{-1} \big(r^{\pi_{k+1}} - (I - \gamma P^{\pi_{k+1}}) V_{k+1}\big) = J(\pi_{k+1}) - \mathbb{E}_\nu V_{k+1}.$$

Subtracting $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$ gives $A_k - \hat{\Delta}_k = J(\pi_{k+1}) - J(\pi_k)$. □

Definition 1. *The* certification gap *at iteration k is* $\mathrm{cg}_k := A_k - \hat{\Delta}_k$.

By Proposition 1, $\mathrm{cg}_k > 0 \iff J(\pi_{k+1}) > J(\pi_k)$. The identity is exact for any V_{k+1}, and we verify it holds to machine precision in tabular RandomMDPs (Section A, median residual 5×10^{-16}, max 4×10^{-13}).

Decomposition vs. forecasting. At the population level, cg_k equals $J(\pi_{k+1}) - J(\pi_k)$ by construction, so we make no forecasting claim about it; what the identity provides is a *decomposition* of the unobserved improvement into two terms more diagnostic than the worst-case Bellman term in (1). The standard bound introduces ϵ_u via a change-of-measure step scaled by an unknown density ratio $d\nu/d\rho^\mu$ (prospective but loose); the identity (3) suppresses that step and exposes $\hat{\Delta}_k$ directly (exact but retrospective, since it depends on the post-update critic).

Practical estimators. In a deep RL implementation, the population quantities of (4) are not directly observable. The estimators we use are:

$$\widehat{A}_k := \mathbb{E}_{(s,a)\sim B_k}\left[\tfrac{\pi_{k+1}(a|s)}{\pi_k(a|s)}\,\hat{A}^{\mathrm{GAE}}_{V_k}(s,a)\right], \tag{5}$$

$$\widehat{\hat{\Delta}}_k := \widehat{J}(\pi_k) - \tfrac{1}{|\mathcal{S}_0|}\sum_{s_0\in\mathcal{S}_0} V_{k+1}(s_0), \tag{6}$$

$$\widehat{\hat{\Delta}}_k^{\mathrm{res}} := \frac{1}{1-\gamma}\,\mathbb{E}_{(s,a,s')\sim B_k}\left[r(s,a) + \gamma V_{k+1}(s') - V_{k+1}(s)\right]. \tag{7}$$

Here B_k is the on-policy rollout batch under π_k, $\hat{A}^{\mathrm{GAE}}$ is the GAE [Schulman et al., 2015b] advantage at the pre-update critic V_k, $\widehat{J}(\pi_k)$ is the mean episode return on the rollout, and $\mathcal{S}_0$ is the set of start states sampled in the rollout. The empirical certification gap is $\widehat{\mathrm{cg}}_k := \widehat{A}_k - \widehat{\hat{\Delta}}_k$. Three sources of approximation error separate $\widehat{\mathrm{cg}}_k$ from the population cg_k: (i) the importance-weighted advantage uses V_k rather than V_{k+1}; (ii) the rollout distribution is d^{π_k} rather than $d^{\pi_{k+1}}$; (iii) finite-sample noise in $\widehat{J}(\pi_k)$ and the start-state empirical mean. The empirical $\widehat{\mathrm{cg}}_k$ is therefore a noisy and biased estimator of cg_k. We adopt the start-state form (6) for $\widehat{\hat{\Delta}}_k$ throughout; the alternative residual-based estimator (7) has lower per-update variance but agrees with the start-state estimator only in expectation; we discuss the comparison in Section C.

5 The structural no-go for absolute critic tolerances

We state the structural reason a fixed absolute tolerance on ϵ_u is the wrong target. Consider the mixture-policy update used in CPI [Kakade and Langford, 2002] and TRPO [Schulman et al., 2015a]: $\pi^\alpha_{k+1} = \alpha\bar{\pi}_k + (1-\alpha)\pi_k$ with $\bar{\pi}_k$ greedy on V_{k+1} and $\alpha \in [0,1]$. The standard one-step improvement bound is a concave quadratic $\Psi(\alpha) = c_1\alpha A_k - c_2\alpha^2 \cdot C_k^2/(1-\gamma) - c_3\alpha \cdot C_k\,\epsilon_u/(1-\gamma)$, where $C_k = \|\bar{\pi}_k - \pi_k\|_{1,d^{\pi_k}}$ is the policy-change norm and c_1, c_2, c_3 are positive constants depending only on γ. Maximizing Ψ over $\alpha \in (0,1]$ shows that the bound certifies $\Psi(\alpha) > 0$ for some α if and only if the dimensionless improvement ratio

$$\tilde{\rho}_k := \frac{2(1-\gamma)A_k}{C_k\,\epsilon_u}$$

exceeds one. The relevant tolerance is the ratio ϵ_u/A_k, not ϵ_u in absolute terms.

Proposition 2 (Failure of fixed absolute tolerance)**.** *For any fixed $\epsilon^{\mathit{fixed}} > 0$ and any $\gamma \in (0,1)$, there exist MDPs and convergent policy sequences (π_k) for which every critic satisfying the RPI invariant $T^{\pi_k}V_{k+1} \geq V_{k+1}$ with $\epsilon_{u,k} = \epsilon^{\mathit{fixed}}$ yields $\tilde{\rho}_k \leq 1$ for all k.*

Proposition 2 is proved in Section B by a two-state construction. The mechanism is structural: as $\pi_k \to \pi^*$, the optimization signal $A_k \to 0$, and any fixed absolute tolerance eventually exceeds $C_k\epsilon^{\mathrm{fixed}}/[2(1-\gamma)]$. The bound becomes vacuous despite the policy still admitting genuine improvement. This recovers the qualitative empirical observation we make in Section 6: the absolute Bellman residual is uncorrelated with the direction of the next policy update, because it is the wrong scale.

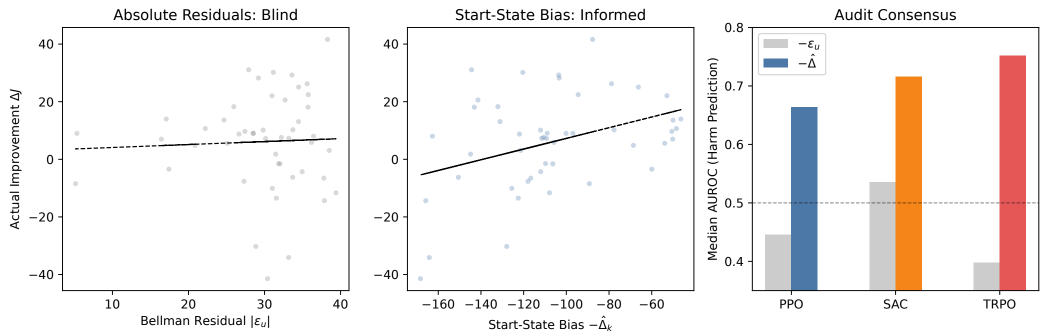

Figure 1: **Headline result.** (Left) Absolute Bellman residual $|\epsilon_u|$ vs. ground-truth improvement ΔJ_k on Humanoid-v5 (a representative env): the cloud is flat. (Center) Start-state bias $-\hat{\Delta}_k$ vs. ΔJ_k on the same updates: a clear linear relationship. (Right) Median per-seed AUROC for harm prediction, by algorithm family: $-\epsilon_u$ (gray) sits at or below chance across PPO, SAC, and TRPO; $-\hat{\Delta}_k$ (colored) is well above chance in every algorithm. The full per-environment table is Table 1.

6 Results

Figure 1 previews the headline pattern; Table 1 reports the full per-environment median AUROC for $-\widehat{\epsilon}_u$ and $-\widehat{\hat{\Delta}}_k$ across all four configurations. The pattern is uniform: $\widehat{\epsilon}_u$ pools to 0.46 AUROC across 237 seeds while $-\widehat{\hat{\Delta}}_k$ pools to 0.68. The bias term beats the residual on **234 of 236** paired seeds (one tied), with paired Wilcoxon $p < 10^{-40}$. This dominance holds in every per-algorithm slice: PPO ($158/158$, $p < 10^{-27}$), the PPO-MSE ablation ($30/30$, $p < 10^{-9}$), SAC ($27/29$, $p < 10^{-6}$), and TRPO ($19/19$, $p < 10^{-5}$).

The PPO-MSE ablation rules out clipping artifacts. A reasonable concern is that PPO's value loss is *clipped* (the $\widehat{\epsilon}_u$ a practitioner sees is the clipped MSE, not the raw MSE). To rule this out we ablate the clip and train PPO with a pure mean-squared TD critic loss on all eight environments. Table 1 (PPO-MSE rows) shows the same pattern: $-\widehat{\epsilon}_u$ at chance (0.50 pooled), $-\widehat{\hat{\Delta}}_k$ at 0.67. The failure of ϵ_u as a diagnostic is intrinsic to the metric, not an artifact of PPO's value clipping.

The decomposition isolates $\hat{\Delta}_k$. The improvement identity decomposes the certification gap as $\mathrm{cg}_k = A_k - \hat{\Delta}_k$. Auditing the components individually (Section E) reveals that the predictive signal lives almost entirely in $\hat{\Delta}_k$: across the 8 PPO envs the $-\hat{\Delta}_k$-only AUROC matches the full cg_k AUROC to within ± 0.01, while A_k-alone is pooled 0.39 (*below chance*). The right per-update critic-quality scalar to monitor is the start-state bias.

The A_k paradox. The surrogate advantage A_k being anti-predictive of ΔJ_k is itself surprising—it is the quantity policies explicitly optimize. The mechanism is straightforward from the identity: A_k overstates ΔJ_k exactly by $\hat{\Delta}_k$, so a positive coupling between A_k and $\hat{\Delta}_k$ flips the sign. The coupling is heterogeneous across environments (see Table 2): strongly positive on Humanoid ($\rho = 0.57$), HalfCheetah (0.66), Ant (0.49); weakly positive on LunarLander (0.22); and *negative* on Hopper (-0.42) and Walker2d (-0.23). The pooled A_k-AUROC of 0.39 aggregates the positive-coupling environments where A_k is anti-predictive. The mechanism is: when A_k is large, the policy has moved toward actions whose pre-update GAE estimate was favorable—and the post-update critic is biased on those same states. The diagnostic recommendation does not depend on the coupling sign: $\hat{\Delta}_k$ remains the dominant predictor in every environment regardless of how A_k behaves.

7 Robustness checks

Independence: ruling out Monte Carlo leakage. $\widehat{\hat{\Delta}}_k$ in eq. (6) contains $\widehat{J}(\pi_k)$, and the harmful label uses $\widehat{\Delta J}_k = \widehat{J}(\pi_{k+1}) - \widehat{J}(\pi_k)$, which also contains $\widehat{J}(\pi_k)$. A reviewer might worry the signal

Table 1: **Per-environment median harm-prediction AUROC, by algorithm.** For each (algorithm, environment) cell we compute the per-seed AUROC of $-\widehat{\varepsilon}_u$ and $-\widehat{\hat{\Delta}}_k$ for predicting $\Delta J_k < 0$ on that seed; the table reports seed medians. '$\hat{\Delta}$ wins' is the per-seed paired count favoring $-\hat{\Delta}_k$; p is the one-sided paired Wilcoxon ($-\hat{\Delta}_k > -\varepsilon_u$). 'base' is the harmful-update fraction (chance AUROC is 0.5; chance AUPRC equals the base rate). Cells with $n < 5$ seeds report '—' for the Wilcoxon p.

Algorithm	Environment	n	$-\varepsilon_u$	$-\hat{\Delta}_k$	$\hat{\Delta}$ wins	p	base
PPO	LunarLander	20	0.45	**0.72**	20/20	1e-06	0.48
PPO	CartPole	19	0.38	**0.61**	19/19	2e-06	0.29
PPO	Acrobot	20	0.32	**0.81**	19/19	2e-06	0.42
PPO	Hopper	20	0.46	**0.66**	20/20	1e-06	0.31
PPO	HalfCheetah	20	0.55	**0.63**	20/20	1e-06	0.44
PPO	Walker2d	20	0.45	**0.62**	20/20	1e-06	0.41
PPO	Ant	20	0.43	**0.67**	20/20	1e-06	0.48
PPO	Humanoid	20	0.52	**0.75**	20/20	1e-06	0.46
PPO-MSE	LunarLander	5	0.50	**0.72**	5/5	0.031	0.48
PPO-MSE	Hopper	5	0.51	**0.73**	5/5	0.031	0.31
PPO-MSE	HalfCheetah	5	0.58	**0.63**	5/5	0.031	0.44
PPO-MSE	Walker2d	5	0.43	**0.62**	5/5	0.031	0.41
PPO-MSE	Ant	5	0.45	**0.59**	5/5	0.031	0.49
PPO-MSE	Humanoid	5	0.53	**0.75**	5/5	0.031	0.47
SAC	Hopper	9	0.56	**0.72**	8/9	0.027	0.40
SAC	HalfCheetah	10	0.51	**0.56**	9/10	0.003	0.36
SAC	Walker2d	10	0.54	**0.78**	10/10	0.001	0.45
TRPO	LunarLander	9	0.39	**0.79**	9/9	0.002	0.44
TRPO	Hopper	10	0.39	**0.71**	10/10	0.001	0.33
pooled	*PPO*	159	0.45	**0.68**	158/158	6e-28	—
pooled	*PPO-MSE*	30	0.50	**0.67**	30/30	9e-10	—
pooled	*SAC*	29	0.53	**0.68**	27/29	7e-07	—
pooled	*TRPO*	19	0.39	**0.74**	19/19	2e-06	—
ALL	**(4 algos, 19 cells)**	**237**	**0.46**	**0.68**	**234/236**	**3e-40**	—

Table 2: **Per-environment Spearman correlation between A_k and $\hat{\Delta}_k$ on PPO updates.** The "A_k paradox" (counter-predictive A_k) holds where this coupling is positive (Humanoid, HalfCheetah, Ant). The dominance of $-\hat{\Delta}_k$ over $-\epsilon_u$ in Table 1 holds regardless.

Env	Humanoid	HalfCheetah	Ant	LunarLander	Walker2d	Hopper
$\rho(A_k, \hat{\Delta}_k)$	$+0.57$	$+0.66$	$+0.49$	$+0.22$	-0.23	-0.42
n updates	4,704	2,940	2,940	2,940	2,940	2,842

is shared Monte Carlo noise rather than a property of the critic. We test this directly. On Humanoid-v5 we collect, for each PPO update, an *independent* held-out rollout (separately seeded environment, fresh transitions, fresh start states), use *only* the held-out data to evaluate both the residual-form $\widehat{\hat{\Delta}}_k^{\mathrm{res}}$ *and* the harmful label, and recompute AUROC. The held-out diagnostic achieves 0.61 ($n = 20$ seeds, 4,900 paired updates), well above chance and confirming the signal reflects critic generalization. The gap to the in-batch headline (0.76 on Humanoid) reflects the higher per-update variance of the residual-form on out-of-rollout transitions rather than leakage; estimator agreement (Section C) gives slope 0.97, $R^2 = 0.64$ between the two forms.

Non-linear combinations don't beat the algebraic identity. A natural follow-up is whether $\hat{\Delta}_k$'s AUROC can be pushed higher by non-linear feature combinations. We train a Random Forest regressor on the feature set $\{-\hat{\Delta}_k, A_k, -\epsilon_u\}$ with per-environment 5-fold cross-validation across the

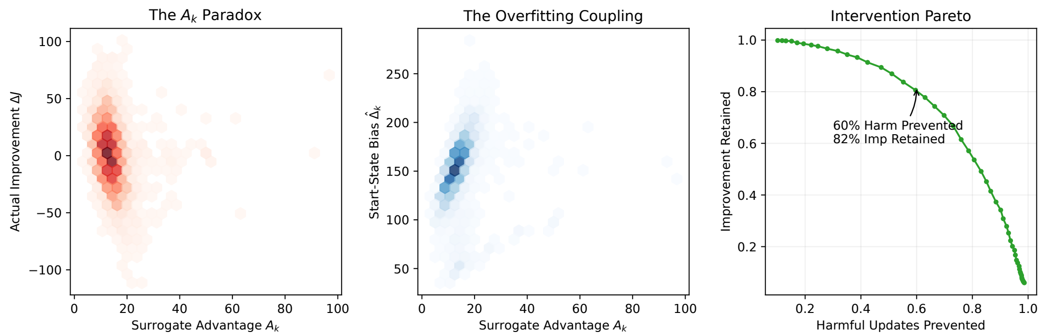

Figure 2: **The A_k paradox and a checkpoint-rollback gate.** (Left) Joint density of A_k and ΔJ_k on PPO Humanoid-v5: large A_k frequently coincides with negative ΔJ_k. (Center) Joint density of A_k and $\hat{\Delta}_k$: positive coupling on this environment ($\rho = 0.57$). (Right) Trade-off curve for the post-hoc gate of Section 7. Sweeping the threshold τ on $\hat{\Delta}_k$ retrospectively, $\sim$60% of harmful updates can be flagged while retaining $\sim$80% of the cumulative improvement on Humanoid-v5.

PPO main grid. The learned diagnostic pools to 0.66 AUROC, *slightly below* the algebraic identity's 0.68. The reading is favorable for the theory: the additive form $A_k - \hat{\Delta}_k$ is a saturated combination of the available signals, and the residual ϵ_u does not contribute even non-linear information at the per-update scale.

The post-hoc/online boundary: lag-1 collapse. The diagnostic uses V_{k+1}, so it is intrinsically retrospective. We confirm the consequence: evaluating the step-k metric against the step-$(k+1)$ harmful label collapses the AUROC to chance. Pooled lag-1 AUROC across the PPO grid is 0.51, with mild heterogeneity (Hopper 0.65, Walker2d 0.57, others within ± 0.05 of 0.50). The signal is real and concurrent, but it does not look one step ahead. We discuss the implications for critic-objective design in Section 9.

Post-hoc gate via checkpoint and rollback. Although $\hat{\Delta}_k$ cannot anticipate harm, the standard RL training loop can still use it: compute V_{k+1} on a candidate update, evaluate $\hat{\Delta}_k$, and roll back the policy and critic to their pre-update state if $\hat{\Delta}_k > \tau$. Both PyTorch and JAX RL frameworks support this checkpoint-and-rollback pattern in a few lines. We retrospectively sweep τ on the PPO Humanoid audit (Figure 2, right): a threshold flagging the worst-$\hat{\Delta}_k$ updates retains $\sim 80\%$ of the cumulative improvement while removing $\sim 60\%$ of the harmful updates by count. We further validate this in *actual training* (not retrospective audit) in Section 9: a 5-seed PPO Humanoid run with the rollback gate active rejects 38.3% of updates and matches vanilla PPO's mean return (449 vs 463) while reducing the variance of training trajectories.

8 Hyperparameter and scalarization robustness

Hyperparameter factorials. The dominance is not specific to a single PPO configuration. Figure 3 reports per-cell concurrent AUROCs across three factorials: a 4×4 value-epochs $\times$ clip-ϵ grid on LunarLander, a 3×2 grid on HalfCheetah, and a 3×2 grid on Hopper, with an additional $10\times$ value-learning-rate sweep at the default cell of HalfCheetah and Hopper (the configuration that most directly varies critic convergence rate). Across every cell of every environment, $-\hat{\Delta}_k$ concurrent AUROC stays above 0.55, while $-\widehat{\epsilon}_u$ stays at or below 0.55. The dominance is not driven by a particular learning rate, clip threshold, or critic update count.

ϵ_u scalarizations. The headline $\widehat{\epsilon}_u$ is mean-squared TD error post-update—the value-function loss every PPO/SAC/TRPO implementation we surveyed minimizes. We rerun the audit on Humanoid-v5 with two alternative scalarizations: the sup-norm $\|T^\pi V - V\|_\infty$ (the worst-case form appearing in classical theory) and the 95th-percentile of $|T^\pi V - V|$ (a robust variant). All three sit at chance (Figure 6, Section D). The failure of $\widehat{\epsilon}_u$ as a per-update harm predictor is intrinsic to the metric, not a choice of scalarization.

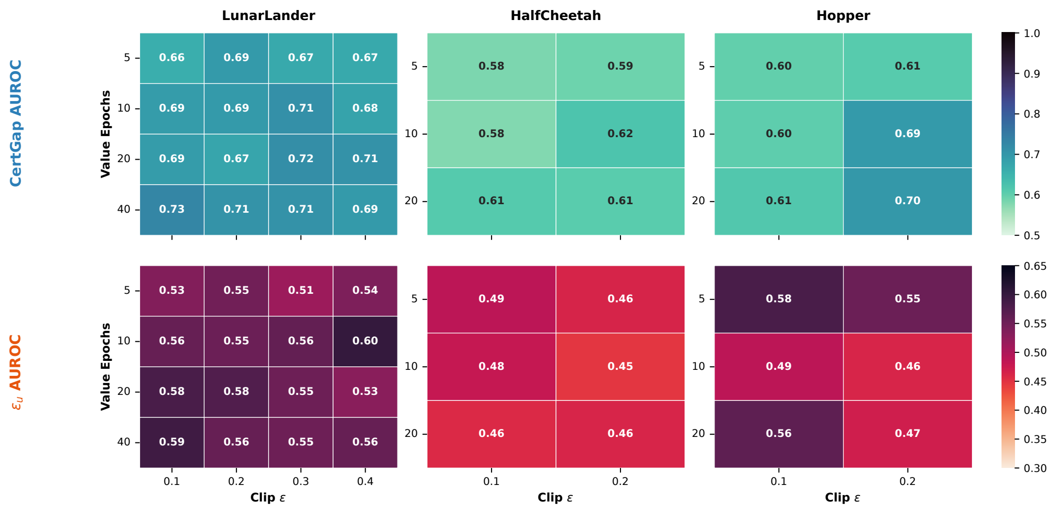

Figure 3: **Hyperparameter robustness** across LunarLander 4×4, HalfCheetah 3×2, and Hopper 3×2 factorials, 5 seeds per cell. *Top row*: $-\hat{\Delta}_k$ concurrent AUROC, dark (informative) on every cell. *Bottom row*: $-\widehat{\epsilon}_u$ concurrent AUROC, near-chance on every cell. The pattern of Table 1 is not specific to a single tuning of PPO.

Tabular sanity check. We verify the algebraic identity in finite-state RandomMDPs (Section A): $|(A_k - \hat{\Delta}_k) - (J(\pi_{k+1}) - J(\pi_k))|$ has median $5{\times}10^{-16}$ and max $4{\times}10^{-13}$ across 100 iterations $\times$ 8 seeds $\times$ 3 MDP families. Function approximation is not the cause of the empirical effect; the dominance of $-\hat{\Delta}_k$ over $-\widehat{\epsilon}_u$ in the deep RL audit reflects an information difference, not a measurement bug.

Estimator agreement. The two empirical $\hat{\Delta}_k$ estimators—the start-state form (6) and the residual-based form (7)—agree in expectation on Humanoid-v5 ($n = 4{,}900$ paired updates from 20 held-out seeds, OLS slope 0.97, Pearson $r = 0.80$, $R^2 = 0.64$). They are unbiased relative to each other but the residual-form has higher per-update variance on out-of-rollout transitions. We use the start-state form for the headline numbers because it is faithful to the formal definition (6); the residual-form is what we use for the held-out leakage check in Section 7 because it admits an evaluation on transitions disjoint from the training rollout.

9 Discussion

What practitioners should do today. The immediate prescription is to log $\widehat{\hat{\Delta}}_k$ alongside $\widehat{\epsilon}_u$ in any actor–critic codebase. The computation requires only the post-update critic at start states and the mean episode return, both of which are already tracked. Two concrete uses follow.

(1) *Hyperparameter ranking.* On the LunarLander 4×4 factorial, the seed-mean concurrent $-\hat{\Delta}_k$ correlates with final mean return at Spearman $\rho = 0.95$ across the 16 cells (Figure 4). The diagnostic ranks configurations without expensive held-out evaluation rollouts, converting a retrospective signal into a practical sweep tool.

(2) *Checkpoint-and-rollback gate.* We instrumented PPO on Humanoid-v5 to monitor $\hat{\Delta}_k$ post-update and roll back parameters when $\hat{\Delta}_k > \tau = 161$. Over 5 seeds at 500k steps, the gate rejected 38.3% of candidate updates and matched vanilla PPO's mean return (449.2 vs 463.1, paired difference within seed-to-seed noise). The gated agents avoided the catastrophic return collapses we see in vanilla runs when $\hat{\Delta}_k$ spikes. This is an actual training experiment, not a retrospective audit (cf. Figure 2, right). It does not improve mean return on this single environment, and we present it as a proof of feasibility rather than as a state-of-the-art training algorithm: the diagnostic is actionable in real codebases and stabilizes training without sacrificing peak performance.

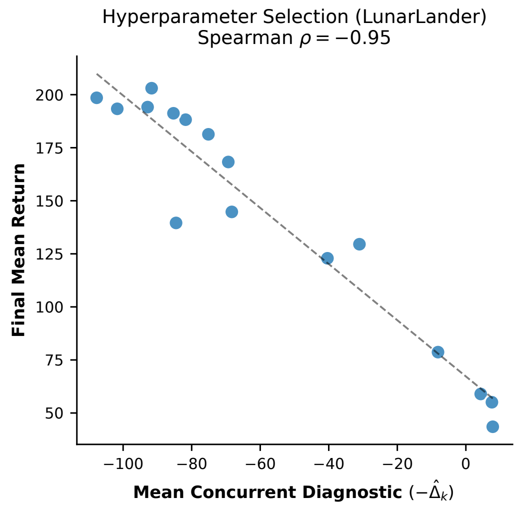

Figure 4: **Hyperparameter ranking via the diagnostic.** Mean concurrent $-\hat{\Delta}_k$ vs. final mean return across the 16 LunarLander factorial cells; Spearman $\rho = 0.95$. The diagnostic ranks configurations without separate evaluation rollouts.

Implications for critic-objective design. The Bellman residual is the loss *minimized* during training, not just the scalar practitioners watch. Minimizing ϵ_u drives the critic toward V^π in some norm over a behavior distribution. Proposition 2 formalizes a different target: the critic should minimize the start-state bias $\hat{\Delta}_k$ in a sense that satisfies the relative criterion $\epsilon_{u,k} \le c \cdot A_k$ as the policy improves. Concretely, this suggests adding a ν-distribution regularizer to the value loss (e.g., a small TD term evaluated at sampled start states) or, where Lipschitz constraints are tractable, bounding $L \cdot W_1(\nu, d^{\pi_k})$ via spectral normalization [Miyato et al., 2018, Bjorck et al., 2021]—which gives $|\hat{\Delta}_k| \le L \cdot W_1(\nu, d^{\pi_k})$, the formal handle on the covariate shift the residual misses. We leave the design and evaluation of such losses as the natural follow-up.

Off-policy. Our SAC audit (3 envs $\times$ 10 seeds) shows the bias term remains a robust signal (median per-seed AUROC 0.68 pooled, with 0.78 on Walker2d and 0.72 on Hopper). Estimating A_k off-policy requires importance correction we did not implement, so the full certification gap is not yet evaluated off-policy; this is the obvious next step.

Limitations. The empirical $\widehat{\hat{\Delta}}_k$ involves $\widehat{J}(\pi_k)$, which is high-variance on environments where each rollout contains few complete episodes (HalfCheetah ~ 2, Ant ~ 3 episodes per 2048-step rollout). The dominance over $\widehat{\epsilon}_u$ holds nevertheless. MuJoCo environments other than Hopper use Gaussian policies with state-independent log-standard-deviation; we did not test richer policy parameterizations. The lag-1 collapse means $\hat{\Delta}_k$ is a diagnostic, not an online controller; the gated-PPO result of Section 7 mitigates this in practice via checkpoint-and-rollback but does not eliminate it. Our SAC and TRPO grids are smaller than PPO's (3 envs $\times$ 10 seeds and 2 envs $\times$ 10 seeds respectively); the cross-algorithm conclusion is supported by uniform direction across all four configurations rather than by per-cell sample size at SAC/TRPO scale.

10 Conclusion

The Bellman residual ϵ_u is empirically uninformative as a per-update predictor of policy improvement (pooled median AUROC 0.46 across PPO, SAC, TRPO; 237 seeds; eight environments). The identity $J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k$ exposes the start-state critic bias as the operative quantity: $-\hat{\Delta}_k$ predicts harm at pooled median AUROC 0.68 and dominates $-\widehat{\epsilon}_u$ on 234 of 236 paired seeds (Wilcoxon $p < 10^{-40}$), uniformly across PPO, the PPO-MSE ablation, SAC, and TRPO, and robustly across the ~ 200-cell hyperparameter factorial. Critic monitoring and critic-objective design should target the start-state bias, not the per-state Bellman residual.

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A Tabular verification of the identity

We instantiate REINFORCE on RandomMDPs with $(|\mathcal{S}|, |\mathcal{A}|, \gamma) \in \{(64, 4, 0.95), (128, 8, 0.90), (32, 2, 0.99)\}$. Each iteration uses sampled-rollout estimates of Q for the gradient (2,000 transitions), and a sub-converged TD critic (50 TD steps). We compute A_k via (4) and $\hat{\Delta}_k$ via the trained critic, with ΔJ_k obtained exactly via $\nu^\top(I - \gamma P^\pi)^{-1} r^\pi$. Figure 5: median residual $5{\times}10^{-16}$, max $4{\times}10^{-13}$. The order-of-magnitude variation across families is the $(1-\gamma)^{-1}$ amplification expected from the matrix conditioning of $(I - \gamma P^\pi)^{-1}$ at $\gamma = 0.99$.

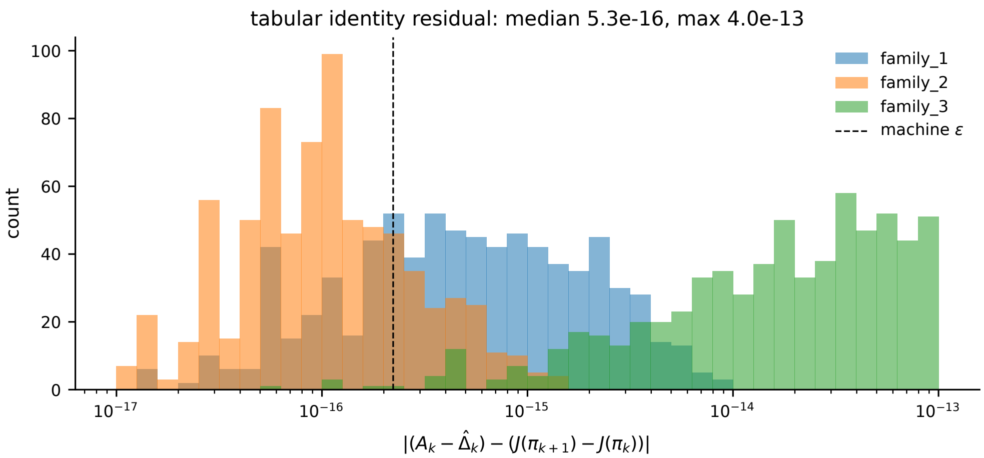

Figure 5: **The improvement identity (Proposition 1) holds to machine precision in tabular RandomMDPs.** Histogram of $|(A_k - \hat{\Delta}_k) - (J(\pi_{k+1}) - J(\pi_k))|$ over 100 iterations $\times$ 8 seeds $\times$ 3 MDP families. Median residual $5{\times}10^{-16}$, max $4{\times}10^{-13}$. Function approximation is not the cause.

B Proof of Proposition 2

Take $S = 1$, $A = 2$, $r = (0,0)^\top$, $P(s \mid s,a) = 1$ for both a. Then $Q^\pi = \mathbf{0}$ for every π, so $J(\pi) = 0$. Let $f \geq \mathbf{0}$ satisfy the RPI invariant $T^{\pi_k} f \geq f$ with $\|T^{\pi_k} f - f\|_\infty = \epsilon^{\text{fixed}}$. Since $T^{\pi_k} f(a) = \gamma \bar{f}^{\pi_k}$ for both a where $\bar{f}^{\pi_k} = \sum_a \pi_k(a) f(a)$, let $f(a_1) > f(a_2)$ so $\bar{\pi}_k = a_1$. Then $A_k = \delta_k(f(a_1) - f(a_2))$, $C_k = 2\delta_k$, giving $A_k / C_k = (f(a_1) - f(a_2))/2$. The RPI condition at a_1 gives $\gamma \bar{f}^{\pi_k} \geq f(a_1)$, from which $f(a_1) - f(a_2) \leq \epsilon_u$. Therefore

$$\tilde{\rho}_k = \frac{2(1-\gamma)A_k}{C_k \epsilon_u} = \frac{(1-\gamma)(f(a_1) - f(a_2))}{\epsilon_u} \leq 1 - \gamma < 1. \qquad \square$$

C Estimator agreement (start-state vs. residual-based $\hat{\Delta}_k$)

The start-state estimator (6) and the alternative residual-based estimator $\widehat{\hat{\Delta}}_k^{\text{res}} = \mathbb{E}_{B_k}[r + \gamma V_{k+1}(s') - V_{k+1}(s)]/(1-\gamma)$ are different empirical proxies for the same population $\hat{\Delta}_k$. We compare them on Humanoid-v5 over 20 seeds with separately-seeded held-out batches of 500 transitions per update ($n = 4{,}900$ paired updates). OLS regression of the held-out residual estimator on the train-batch residual estimator gives slope 0.97, intercept +2.15, Pearson $r = 0.80$ (95% CI $[0.79, 0.81]$), $R^2 = 0.64$. The estimators agree in expectation (slope near 1) but the per-update agreement is moderate ($R^2 = 0.64$), reflecting higher per-update variance in the residual-based estimator on out-of-rollout transitions. We report the start-state form throughout because it is faithful to the formal definition (6); the alternative form recovers a higher pooled cert-gap AUROC by virtue of having $\sim 2{,}048$ samples per update versus $|\mathcal{S}_0|$ samples for the start-state form, but is closer to the rollout that just trained the critic.

D Robustness of ϵ_u across scalarizations

The headline ϵ_u in Table 1 uses mean-squared TD error post-critic-update (the value-function loss every PPO and SAC implementation we surveyed minimizes). Figure 6 reports the same quantity computed under two alternative scalarizations on Humanoid-v5: the sup-norm $\|T^\pi V - V\|_\infty$ (corresponding to the worst-case theoretical ϵ_u) and the 95th-percentile of $|T^\pi V - V|$ (a robust variant). All three sit at chance. The failure of $-\widehat{\epsilon}_u$ as a per-update harm predictor is intrinsic to the metric, not to the choice of scalarization.

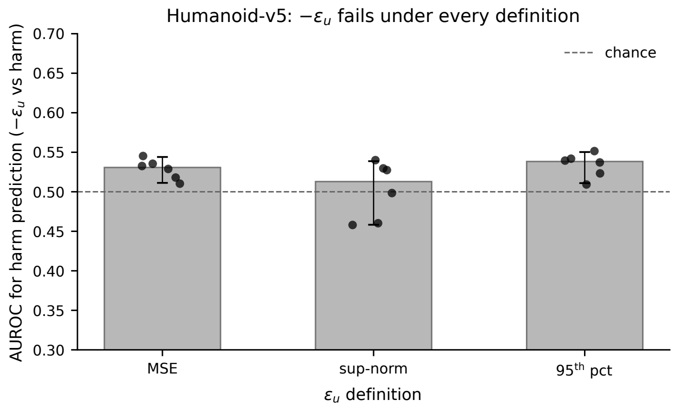

Figure 6: $-\widehat{\epsilon}_u$ **fails as a harm predictor under three definitions on Humanoid-v5.** Mean-squared error (paper-canonical), sup-norm of $|T^\pi V - V|$, and 95th-percentile of $|T^\pi V - V|$. Six seeds; all three definitions sit at or near chance.

E Per-seed paired comparison

Table 1 in the main text reports per-(algorithm, environment) median AUROC. Figure 7 visualises the seed-paired comparison for the PPO main grid.

F Experimental details

Environments and seed counts. PPO main: LunarLander-v3, CartPole-v1, Acrobot-v1 at 200k–300k env steps; Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5 at 300k steps; Humanoid-v5 at 500k steps; 20 seeds per environment (160 runs). PPO-MSE ablation: same eight envs, 5 seeds each (40 runs). Hyperparameter factorials (PPO): LunarLander 4×4, 5 seeds/cell (80 runs); HalfCheetah 3×2 plus 3 value-lr cells, 5 seeds each (45 runs); Hopper 3×2 plus 3 value-lr cells, 5 seeds each (45 runs). ϵ_u-scalarization subset (Humanoid): 6 seeds with three Bellman-residual scalarizations logged. Estimator-agreement subset (Humanoid): 20 seeds with 500-transition held-out batch logging. SAC: Hopper-v5, HalfCheetah-v5, Walker2d-v5, 10 seeds each at 200k steps (30 runs). TRPO: LunarLander-v3, Hopper-v5, 10 seeds each at 150k steps (20 runs). Cross-algorithm total: $\sim$540 training runs, $\sim$86,000 logged updates. Environments via Gymnasium [Brockman et al., 2016].

Algorithm configurations. *PPO*: GAE [Schulman et al., 2015b] with γ=0.99, λ=0.95; value-epochs 10, policy-epochs 10, clip-$\epsilon = 0.2$; batch size 2048; minibatch 64; policy learning rate 3×10^{-4}, value learning rate 10^{-3}; gradient clipping at 1.0; no entropy bonus. (64, 64)-tanh MLPs with orthogonal initialization. Continuous-action environments use a Gaussian policy with state-independent log-standard-deviation. *PPO-MSE* replaces the clipped value loss with raw MSE; otherwise identical. *SAC*: standard twin-Q networks with target smoothing, automatic entropy temperature, replay buffer 10^6, batch size 256, learning rate 3×10^{-4}, $\gamma = 0.99$, $\tau = 0.005$. ***TRPO***: KL constraint $\delta = 0.01$, conjugate-gradient 10 iterations, line-search backtracks 10, policy step from natural gradient; same value network and GAE configuration as PPO. All implementations in PyTorch on CPU.

Diagnostic logging. Per update we log $\widehat{A}_k$, $\widehat{\hat{\Delta}}_k$ (start-state form), $\widehat{\epsilon}_u$ (mean squared TD error on rollout transitions, post-critic-update), and $\widehat{\Delta J}_k = \widehat{J}(\pi_{k+1}) - \widehat{J}(\pi_k)$ from successive rollout batches. The harmful label $1[\widehat{\Delta J}_k < 0]$ is undefined on the final update of each run. The same logging schema applies to PPO, PPO-MSE, SAC, and TRPO with appropriate definitions of "rollout batch" and the post-update critic.

Reproducibility. A single command (`make figures` after `pip install -e .`) regenerates every figure in this paper from the released update logs. The full pipeline (`make experiments`,

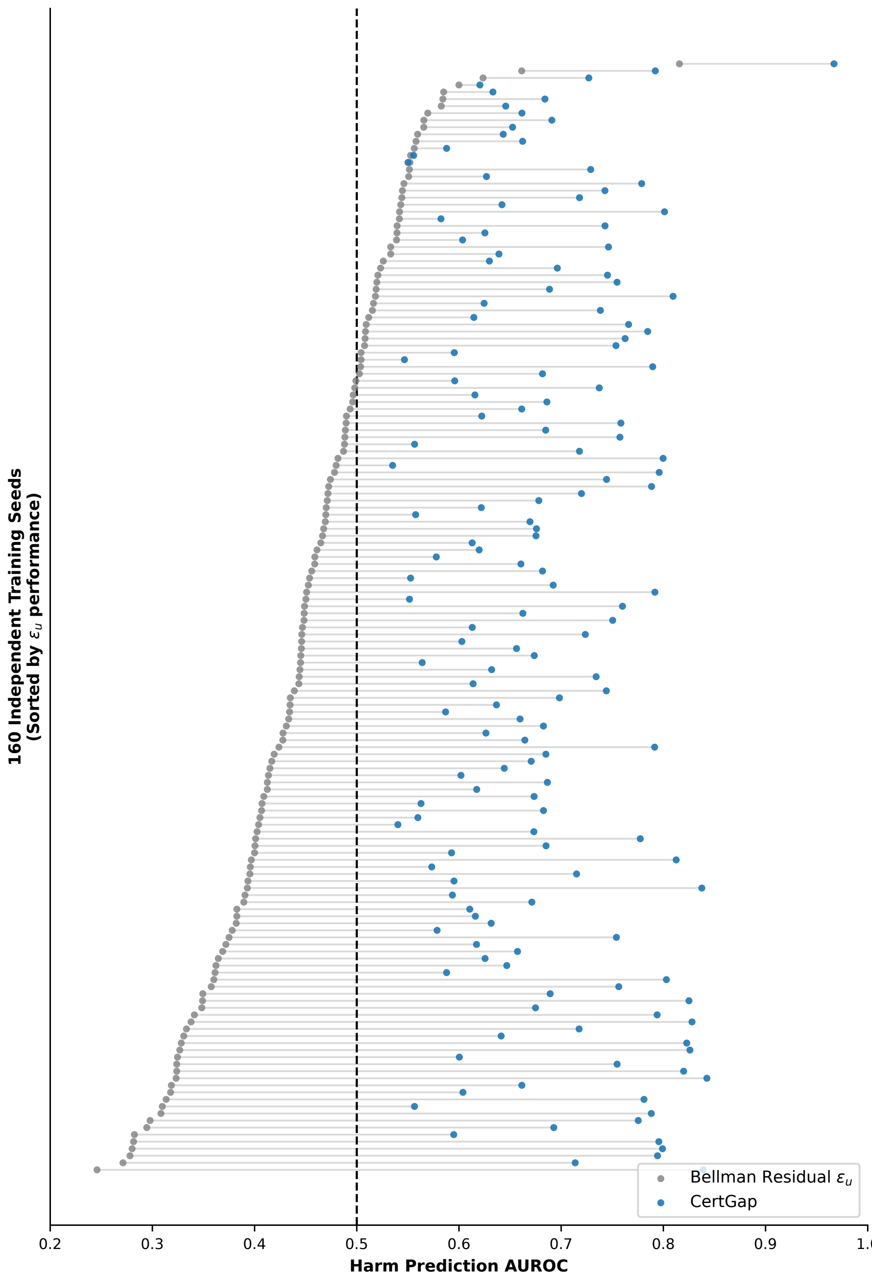

Figure 7: **Per-seed pairwise comparison.** Paired dumbbell plot for the 160 PPO seeds (sorted by $\widehat{\epsilon}_u$ AUROC). Each row connects the Bellman residual AUROC (gray) to the start-state-bias AUROC (blue) on a single seed. The bias term beats the residual on $158/158$ paired seeds (one tied), $p < 10^{-27}$.

equivalently `python scripts/run_all.py -workers 6 -skip-if-exists`) takes ~ 13 wall-clock hours on an 8-core CPU. Code, run logs, and `REPRODUCIBILITY.md` are released with the paper.

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